

## WEEK 5 – PAYMENT TRANSACTION MANAGEMENT SYSTEM

**Secenario:** The **Payment Transaction Management System** is used to store and manage customer payment transaction details. It maintains payment information related to each order.

**PAYMENT\_ENTITY:** The **PAYMENT\_ENTITY** table stores customer payment transaction details.

### Attributes:

- **PAYMENT\_ID** – Unique payment identifier
- **ORDER\_ID** – Order reference
- **PAYMENT\_DATE** – Date of transaction
- **PAYMENT\_MODE** – UPI / Card / Net Banking / Cash
- **PAYMENT\_STATUS** – Success / Failed / Pending
- **TRANSACTION\_AMOUNT** – Paid amount

```
SQL> CREATE TABLE PAYMENT_ENTITY (  
2     PAYMENT_ID INT,  
3     ORDER_ID INT,  
4     PAYMENT_DATE DATE,  
5     PAYMENT_MODE VARCHAR2(20),  
6     PAYMENT_STATUS VARCHAR2(20),  
7     TRANSACTION_AMOUNT NUMBER  
8 );  
  
Table created.
```

### Establish Relationships and Constraints :

**One Order** can have **one Payment Transaction**.

The relationship between **ORDER\_ENTITY** and **PAYMENT\_ENTITY** is **One to One (1 → 1)**.

### PRIMARY KEY:

The following Primary Key is added:

- **PAYMENT\_ID** – Primary Key of **PAYMENT\_ENTITY** table.

A Primary Key uniquely identifies each record in a table.

```
SQL> ALTER TABLE PAYMENT_ENTITY  
2 ADD CONSTRAINT PK_PAYMENT  
3 PRIMARY KEY (PAYMENT_ID);  
  
Table altered.
```

### FOREIGN KEY:

The following Foreign Key is used in the **PAYMENT\_ENTITY** table:

- **ORDER\_ID** references **ORDER\_ENTITY(ORDER\_ID)**

This Foreign Key establishes the relationship between the two tables.

```
SQL> ALTER TABLE PAYMENT_ENTITY
2 ADD CONSTRAINT FK_PAYMENT_ORDER
3 FOREIGN KEY (ORDER_ID)
4 REFERENCES ORDER_ENTITY(ORDER_ID);
```

Table altered.

### NOT NULL Constraint:

NOT NULL ensures that a column cannot contain an empty (NULL) value.

✚ PAYMENT\_MODE  
✚ PAYMENT\_STATUS

```
SQL> ALTER TABLE PAYMENT_ENTITY
2 MODIFY PAYMENT_MODE NOT NULL;
```

Table altered.

```
SQL> ALTER TABLE PAYMENT_ENTITY
2 MODIFY PAYMENT_STATUS NOT NULL;
```

Table altered.

### CHECK Constraint:

CHECK validates that the values in a column satisfy a specified condition.

- TRANSACTION\_AMOUNT > 0 – Ensures the transaction amount is positive.

```
SQL> ALTER TABLE PAYMENT_ENTITY
2 ADD CONSTRAINT CHK_TRANSACTION_AMOUNT
3 CHECK (TRANSACTION_AMOUNT > 0);
```

Table altered.

### INSERT VALUES FOR PAYMENT ENTITY:

```
SQL> INSERT INTO PAYMENT_ENTITY
2 VALUES ('PAY001', 1001, TO_DATE('20-09-2026','DD-MM-YYYY'), 'UPI', 'SUCCESS', 1500);
1 row created.

SQL> INSERT INTO PAYMENT_ENTITY
2 VALUES ('PAY002', 1002, TO_DATE('21-09-2026','DD-MM-YYYY'), 'CARD', 'SUCCESS', 2200);
1 row created.

SQL> INSERT INTO PAYMENT_ENTITY
2 VALUES ('PAY003', 1003, TO_DATE('22-09-2026','DD-MM-YYYY'), 'NET BANKING', 'SUCCESS', 1800);
1 row created.

SQL> INSERT INTO PAYMENT_ENTITY
2 VALUES ('PAY004', 1004, TO_DATE('23-09-2026','DD-MM-YYYY'), 'UPI', 'SUCCESS', 2500);
1 row created.

SQL> INSERT INTO PAYMENT_ENTITY
2 VALUES ('PAY005', 1005, TO_DATE('24-09-2026','DD-MM-YYYY'), 'CARD', 'PENDING', 1200);
1 row created.

SQL> INSERT INTO PAYMENT_ENTITY
2 VALUES ('PAY006', 1006, TO_DATE('25-09-2026','DD-MM-YYYY'), 'UPI', 'SUCCESS', 1700);
1 row created.

SQL> INSERT INTO PAYMENT_ENTITY
2 VALUES ('PAY007', 1007, TO_DATE('26-09-2026','DD-MM-YYYY'), 'CARD', 'SUCCESS', 1900);
1 row created.

SQL> INSERT INTO PAYMENT_ENTITY
2 VALUES ('PAY008', 1008, TO_DATE('27-09-2026','DD-MM-YYYY'), 'NET BANKING', 'SUCCESS', 2300);
1 row created.

SQL> INSERT INTO PAYMENT_ENTITY
2 VALUES ('PAY009', 1009, TO_DATE('28-09-2026','DD-MM-YYYY'), 'UPI', 'SUCCESS', 1400);
1 row created.

SQL> INSERT INTO PAYMENT_ENTITY
2 VALUES ('PAY010', 1010, TO_DATE('29-09-2026','DD-MM-YYYY'), 'CARD', 'SUCCESS', 2800);
1 row created.
```

```
SQL> SELECT * FROM PAYMENT_ENTITY;
```

| PAYMENT_ID         | ORDER_ID | PAYMENT_D | PAYMENT_MODE | PAYMENT_STATUS |
|--------------------|----------|-----------|--------------|----------------|
| TRANSACTION_AMOUNT |          |           |              |                |
| PAY001             | 1001     | 20-SEP-26 | UPI          | SUCCESS        |
|                    | 1500     |           |              |                |
| PAY002             | 1002     | 21-SEP-26 | CARD         | SUCCESS        |
|                    | 2200     |           |              |                |
| PAY003             | 1003     | 22-SEP-26 | NET BANKING  | SUCCESS        |
|                    | 1800     |           |              |                |
| PAYMENT_ID         | ORDER_ID | PAYMENT_D | PAYMENT_MODE | PAYMENT_STATUS |
| TRANSACTION_AMOUNT |          |           |              |                |
| PAY004             | 1004     | 23-SEP-26 | UPI          | SUCCESS        |
|                    | 2500     |           |              |                |
| PAY005             | 1005     | 24-SEP-26 | CARD         | PENDING        |
|                    | 1200     |           |              |                |
| PAY006             | 1006     | 25-SEP-26 | UPI          | SUCCESS        |
|                    | 1700     |           |              |                |
| PAYMENT_ID         | ORDER_ID | PAYMENT_D | PAYMENT_MODE | PAYMENT_STATUS |
| TRANSACTION_AMOUNT |          |           |              |                |
| PAY007             | 1007     | 26-SEP-26 | CARD         | SUCCESS        |
|                    | 1900     |           |              |                |
| PAY008             | 1008     | 27-SEP-26 | NET BANKING  | SUCCESS        |
|                    | 2300     |           |              |                |
| PAY009             | 1009     | 28-SEP-26 | UPI          | SUCCESS        |
|                    | 1400     |           |              |                |
| PAYMENT_ID         | ORDER_ID | PAYMENT_D | PAYMENT_MODE | PAYMENT_STATUS |
| TRANSACTION_AMOUNT |          |           |              |                |
| PAY010             | 1010     | 29-SEP-26 | CARD         | SUCCESS        |
|                    | 2800     |           |              |                |

10 rows selected.

## REPORT 1 : Display all successful payments

```
SQL> SELECT *  
2 FROM PAYMENT_ENTITY  
3 WHERE PAYMENT_STATUS = 'SUCCESS';
```

| PAYMENT_ID         | ORDER_ID | PAYMENT_D | PAYMENT_MODE | PAYMENT_STATUS |
|--------------------|----------|-----------|--------------|----------------|
| TRANSACTION_AMOUNT |          |           |              |                |
| PAY001             | 1001     | 20-SEP-26 | UPI          | SUCCESS        |
|                    | 1500     |           |              |                |
| PAY002             | 1002     | 21-SEP-26 | CARD         | SUCCESS        |
|                    | 2200     |           |              |                |
| PAY004             | 1004     | 23-SEP-26 | UPI          | SUCCESS        |
|                    | 2500     |           |              |                |
| PAYMENT_ID         | ORDER_ID | PAYMENT_D | PAYMENT_MODE | PAYMENT_STATUS |
| TRANSACTION_AMOUNT |          |           |              |                |
| PAY006             | 1006     | 25-SEP-26 | UPI          | SUCCESS        |
|                    | 1700     |           |              |                |
| PAY007             | 1007     | 26-SEP-26 | CARD         | SUCCESS        |
|                    | 1900     |           |              |                |
| PAY008             | 1008     | 27-SEP-26 | NET BANKING  | SUCCESS        |
|                    | 2300     |           |              |                |
| PAYMENT_ID         | ORDER_ID | PAYMENT_D | PAYMENT_MODE | PAYMENT_STATUS |
| TRANSACTION_AMOUNT |          |           |              |                |
| PAY009             | 1009     | 28-SEP-26 | UPI          | SUCCESS        |
|                    | 1400     |           |              |                |
| PAY010             | 1010     | 29-SEP-26 | CARD         | SUCCESS        |
|                    | 2800     |           |              |                |

8 rows selected.

## REPORT 2: Failed Transactions

```
SQL> SELECT *
  2 FROM PAYMENT_ENTITY
  3 WHERE PAYMENT_STATUS = 'FAILED';
```

| PAYMENT_ID | ORDER_ID | PAYMENT_D | PAYMENT_MODE | PAYMENT_STATUS | TRANSACTION_AMOUNT |
|------------|----------|-----------|--------------|----------------|--------------------|
| PAY003     | 1003     | 22-SEP-26 | NET BANKING  | FAILED         | 1800               |

### REPORT 3 : Count total successful and failed payments

```
SQL> SELECT PAYMENT_STATUS,
  2 COUNT(*) AS TOTAL_PAYMENTS
  3 FROM PAYMENT_ENTITY
  4 WHERE PAYMENT_STATUS IN ('SUCCESS', 'FAILED')
  5 GROUP BY PAYMENT_STATUS;
```

| PAYMENT_STATUS | TOTAL_PAYMENTS |
|----------------|----------------|
| FAILED         | 1              |
| SUCCESS        | 8              |

### REPORT 4 : Update failed payments after retry

```
SQL> UPDATE PAYMENT_ENTITY
  2 SET PAYMENT_STATUS = 'SUCCESS'
  3 WHERE PAYMENT_ID = 'PAY003'
  4 AND PAYMENT_STATUS = 'FAILED';

1 row updated.
```

### REPORT 5: Identify pending transactions

```
SQL> SELECT *
  2 FROM PAYMENT_ENTITY
  3 WHERE PAYMENT_STATUS = 'PENDING';
```

| PAYMENT_ID | ORDER_ID | PAYMENT_D | PAYMENT_MODE | PAYMENT_STATUS | TRANSACTION_AMOUNT |
|------------|----------|-----------|--------------|----------------|--------------------|
| PAY005     | 1005     | 24-SEP-26 | CARD         | PENDING        | 1200               |